

**THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NJOMBE REGION
FORM SIX PRE-MOCK EXAMINATION**

143/1

MATHEMATICS 1
(For Both School and Private Candidate)

Time 3 Hours **Wednesday, 19th August 2026 a.m**

Instructions

1. This paper consists of **Ten (10)** questions each carrying **Ten (10)** marks.
2. Answer **all** questions.
3. **All** necessary working and answers for each question done must be shown clearly.
4. Mathematical tables and non-programmable calculator may be used.
5. Cellular phones are **not allowed** in the examination room.
6. Write your examination number on **every page** of your answer booklet(s)

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1. (a) Symbolize and hence draw an electrical network
 “It is not true that Rama studies Economics. But Rama studies
 Mathematics or Geography”.
- (b) Use truth table to check if $p \rightarrow (q \wedge r) \equiv (p \rightarrow q) \wedge (p \rightarrow r)$.
- (c) Use the laws of algebra of proposition to simplify

$$[(p \rightarrow q) \wedge \neg q] \rightarrow \neg p$$

2. (a) Solve for x given $\operatorname{arc} \cosh 2x = \operatorname{arc} \sinh x$

(b) If $f(x) = \sinh x \cos x + \sin x \cosh x$

(i) Find a simplified expression of $f'(x)$

(ii) Use the results in (b) (i) to evaluate

$$\int \frac{2}{\tanh x + \tan x} dx$$

3. A TV dealer has stores in Dar es Salaam and Morogoro and retailers in Tanga and Dodoma. The stores have a stock of 45 TV and 40 TV sets respectively while the requirements of the retailers are 25 and 30 TV sets respectively. If the cost of transporting a TV from Dar es Salaam to Tanga is Tsh. 5,000/= and from Dar es Salaam to Dodoma is Tsh. 9,000/=, from Morogoro to Dodoma is Tsh. 3,000/= and Morogoro to Tanga is Tsh. 6,000/=

- (a) How should the TV dealer supply the requested TV sets at minimum cost?
- (b) What is the minimum cost?

4. (a) If for a distribution $\sum (x - 5) = 3$, $\sum (x - 5)^2 = 43$ and the total number of item is 18, find the mean and standard deviation

- (b) The mean weight of 150 students in a certain class is 60 kg. The mean weight of boys is 70 kg and that of girls in the class is 55 kg. Find the number of boys and girls in the class.

- (c) The frequency distribution

x	p	$2p$	$3p$	$4p$	$5p$	$6p$
f	2	1	1	1	1	1

Where p is a positive integer, as a variance of 160. Find the value of p .

5. (a) Use laws of algebra to simplify $(A \cup B) \Delta B'$
- (b) Given $n(A \cap B) = 5$, $n(A \cup B) = 16$, $n(A \cup B)' = 4$ and $n(A - B) = 8$ find (i) $n(A \cup B')$ (ii) $n(A' \cap B)$
- (c) Out of 26 animals in a zoo, 5 animals eat all types of food (grass, meat and bones), 6 animals eat grass and meat only and 4 animals eat meat and bones only, 2 animals eat grass and bones only. The number of animals eating one type of food only is divided equally between the three types of food. Using Venn diagram, find the number of animals eating grass.
6. (a) Given that $f(x) = 4x + 1$ and $g(x) = 6x + k$ Find the value of k if $f \circ g(x) = g \circ f(x)$
- (b) The quadratic function graph passes through the three points $(-3, 4)$, $(-2, 2)$ and $(-4, 2)$. Express the function in the form of $a(x - h)^2 = f(x) - k$
- (c) Draw the graph of the function $f(x) = \frac{x^2 - 2x + 1}{x^3 + x^2 - 2x}$
7. (a) Verify that the roots of $\cos x - xe^x = 0$ lies between 0 and 1 also by secant method to four decimal places. Perform only one iterations.
- (b) Estimate the value of the integral $\int_0^1 xe^{-x^2} dx$ (keep your answers in four (4) decimal places).
- (i) By Simpson's rule using five ordinates
- (ii) By means of the first four terms of the Maclaurin series for e^{-x^2}
- (iii) By analytical method

8. (a) The shortest distance of the point $(p, 2)$ from the line $3x + 4y + 1 = 0$ is $\frac{\sqrt{2}}{5}$ units Find the value of p

(b) Find the secant of the acute angle between the pair of lines

$$y^2 + 2x^2 - 3xy + 5x - y - 12 = 0$$

(c) A bicycle tire has equation $x^2 + y^2 - 4x - 6y + 11 = 0$ find the coordinate of the hub and approximately length of spokes.

9. (a) Prove that derivative of $\sqrt{\frac{1 + \sin x}{1 - \sin x}}$ is $\frac{1}{1 - \sin x}$

(b) Find the derivative of $y = (\tan x + \sec x)^n$ where n is positive integer

(c) A chemical factory wishes to make a cylindrical container of thin metal to hold 126 cm^3 , using the least possible area of a metal. If the outside surface area $A \text{ cm}^2$ and the radius is $r \text{ cm}$, then

(i) Show that

$$A = \frac{2}{r}(\pi r^3 + 126)$$

(ii) Find the required radius and height of the container.

10.(a) Evaluate

$$\int \frac{x^2}{\sqrt{4 - x^2}} dx$$

(b) Show that

$$\int_0^1 \frac{x^2}{(5 - x)} dx = -\frac{11}{2} + 25 \ln \frac{5}{4}$$

(c) Find the length of the curve given by the parametric equations

$$x = 3 \sin 3\theta \text{ and } y = 3 \cos 3\theta, \quad 0 \leq \theta \leq 2\pi$$